

Lab: Action and Decision-Making

Learning Goal: Analyze how agents select actions by balancing exploration and exploitation.

Part 1: Exploration vs. Exploitation

Scenario: You're in a new city for the weekend with three meal opportunities remaining.

Decision	Explore (try somewhere new)	Exploit (go somewhere known)
Dinner tonight		
Lunch tomorrow		
Dinner tomorrow		

For each meal, decide whether you would explore or exploit, and explain your reasoning. How does your remaining time (uncertainty window) affect your decision?

Write your response here

Part 2: Action as Prediction Fulfillment

Learning Goal: Trace the action-as-inference process in specific motor behaviors.

Instructions: For each action, identify: (a) the prediction, (b) the sensory feedback, (c) the prediction error, (d) the motor adjustment.

1. Catching a ball thrown to you

2. Typing a word on a keyboard without looking
3. Walking on an icy sidewalk

Write your response here

Part 3: Expected Free Energy Analysis

Learning Goal: Evaluate actions using the EFE framework.

Scenario: You're deciding between two summer internships:

- **Internship A:** At a company you know well, doing work you're good at. Comfortable but predictable.
- **Internship B:** At a startup in a new field. Uncertain but potentially very rewarding.

Analyze each option using:

1. Pragmatic value (how likely to achieve preferred outcomes?)
2. Epistemic value (how much will you learn / reduce uncertainty?)
3. Overall EFE: Which option has lower expected free energy?

Write your response here

Part 4: When Action Goes Wrong

Learning Goal: Apply the framework to clinical conditions affecting action.

Case analysis: For each condition, explain the disruption in Active Inference terms.

1. **Akinesia in Parkinson's:** A patient wants to walk but can't initiate the movement.

2. **Compulsive hand-washing in OCD:** A patient washes hands 50+ times per day despite knowing they're clean.
3. **Impulsive spending:** A person buys expensive items on impulse and regrets it later.

Write your response here

Part 5: Reflection

In 150 words, discuss: If action is about making the world match predictions, does free will exist? Or are we just biological machines fulfilling our generative model's predictions?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Decision analysis	Exploration vs. exploitation
2	Motor analysis	Action as prediction fulfillment
3	Comparative evaluation	Expected Free Energy components
4	Clinical reasoning	Action disruption in disorders
5	Philosophical reflection	Free will and Active Inference